

KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM agents when to act directly, self-reflect, or consult external knowledge during planning.

Fast thinking

Slow thinking

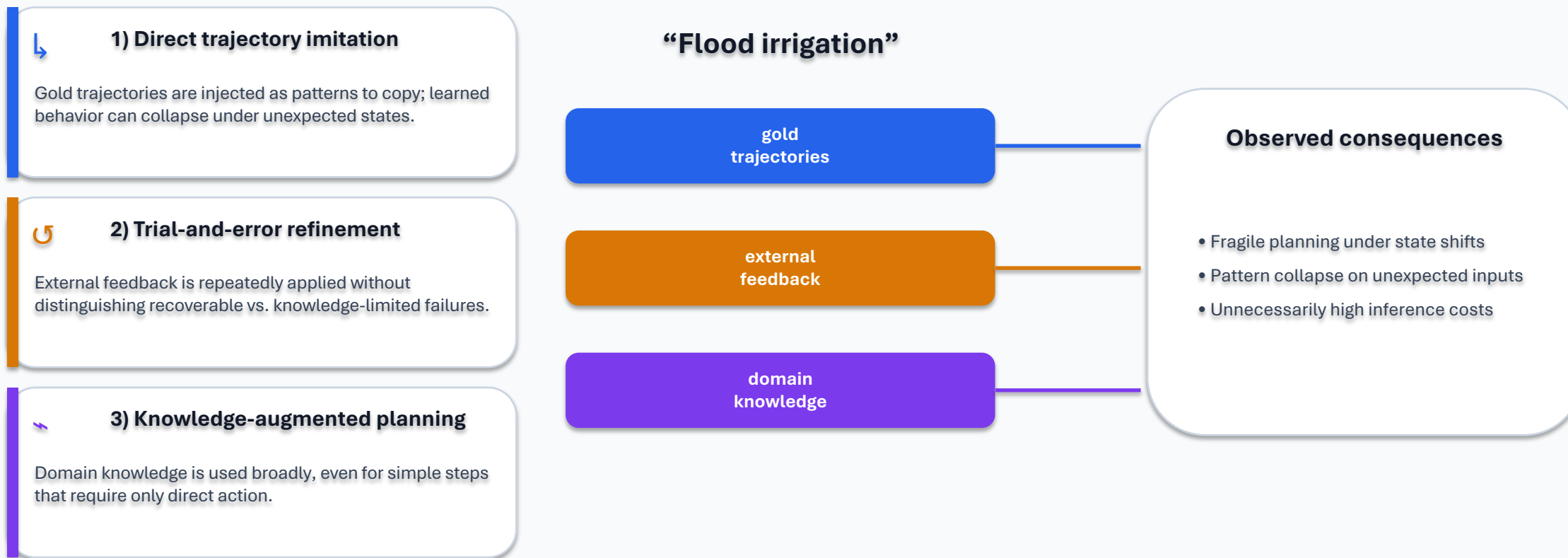
Knowledgeable thinking



SELF-
AWARE
AGENT

Current Agent Training Uses ‘Flood Irrigation’

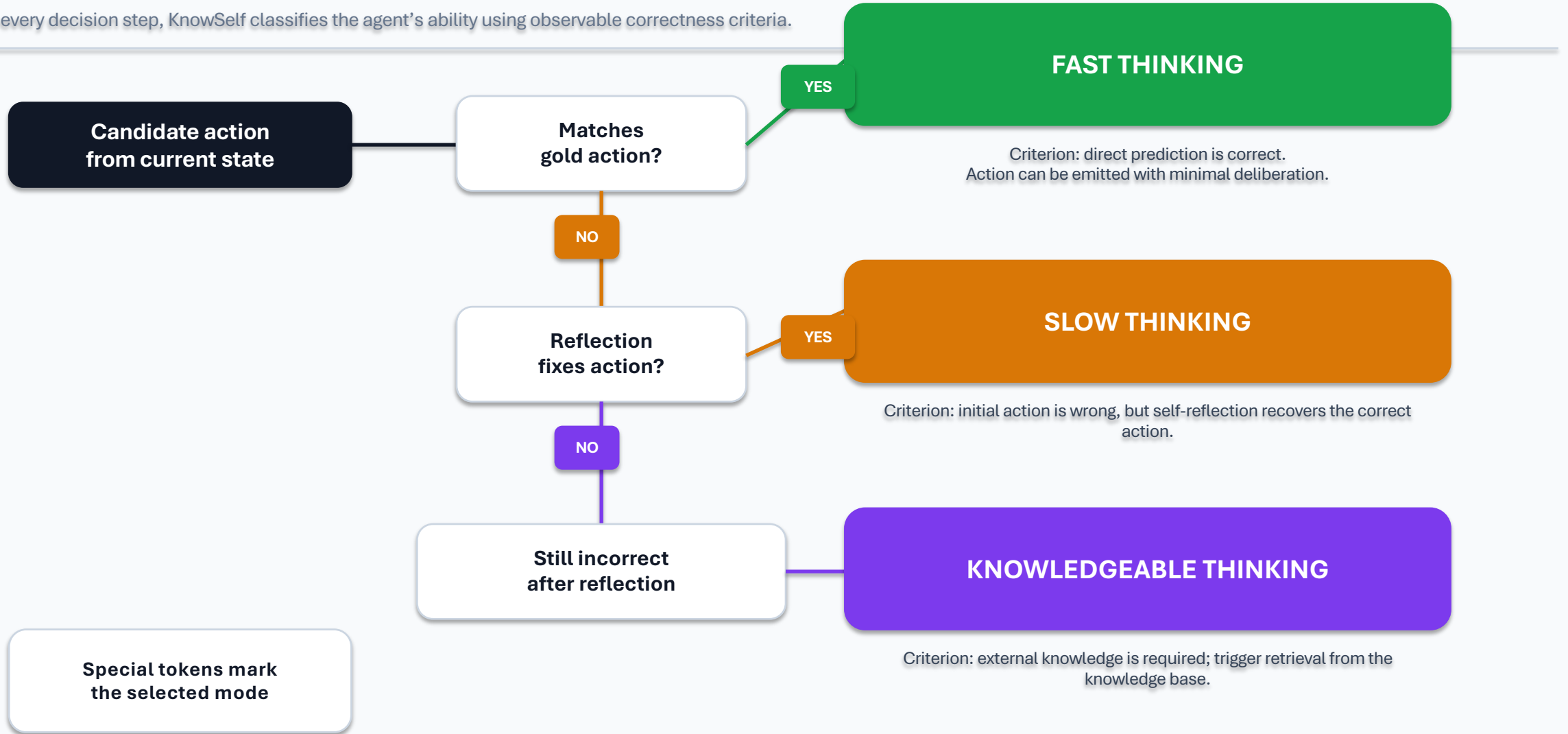
Indiscriminate knowledge/feedback injection ignores whether the agent needs help at a given step.



KnowSelf reframes planning as metacognitive control: assess whether the next step requires direct action, reflection, or external help.

Three Situational Modes: Fast, Slow, Knowledgeable

At every decision step, KnowSelf classifies the agent's ability using observable correctness criteria.



KnowSelf Framework: Data → Training → Inference

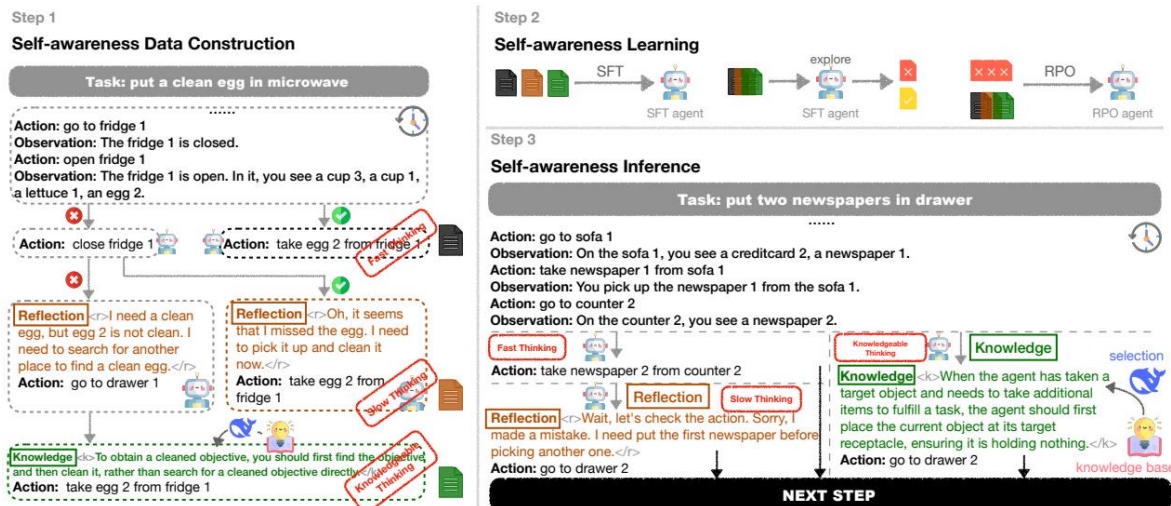
A lightweight pipeline learns when to emit special tokens that control reasoning and knowledge retrieval.

1. Self-awareness data construction 2. Two-stage training 3. Self-aware inference

Agent self-explores environments. Heuristic judgment labels each step as fast, slow, or knowledgeable; special tokens are inserted.

SFT teaches initial self-awareness patterns; Reward-based Policy Optimization refines appropriate token generation.

Generated tokens trigger direct action, reflection, or selective knowledge retrieval during deployment.



Framework figure: data construction, learning, and inference pipeline.

Qualitative Grounding: KnowSelf Recovers Where Strong Models Fail

Two ALFWorld tasks illustrate mode-specific intervention: reflection for recoverable errors; knowledge for unrecoverable ones.

(a) Task: put a hot mug in cabinet	
OpenAI-O1 Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. [Knowledge]< >If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</> Action: take egg 1 from microwave 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. Action: put mug 1 in/on microwave 1 [Reflection]< >Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</> Action: heat mug 1 with microwave 1
(b) Task: put a saltshaker in drawer	
DeepSeek-R1 Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is open. In it, you see a fork 1. Action: put saltshaker 1 in/on drawer 1 [Reflection]< >Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</> Action: close drawer 1 Observation: You close the drawer 1. Action: put saltshaker 1 in/on drawer 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is closed. Action: open drawer 1 Observation: You open the drawer 1. In it, you see a fork 1. [Knowledge]< >When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</> Action: put saltshaker 1 in/on drawer 1

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Task A: “put a hot mug in cabinet”

OpenAI-O1 opens the microwave but incorrectly tries to take an egg. KnowSelf (Llama-8B) emits a Reflection token and realizes the mug should be heated using the open microwave.

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Task B: “put a saltshaker in drawer”

DeepSeek-R1 reflects that the drawer is open and tries to close it first, causing failure. KnowSelf emits a Knowledge token: place the object without removing unrelated items.

Interpretation: KnowSelf’s advantage is not longer reasoning by default; it selects the cheapest effective cognitive operation.

ALFWorld Results: High Accuracy with Selective Knowledge

Success rates (%) across six ALFWorld task types; Know% is the proportion of steps using external knowledge.

Llama-8B KnowSelf: 84.33% All with 15.01% Know

Gemma-2B KnowSelf: 79.85% All with 26.41% Know

100% knowledge methods often underperform

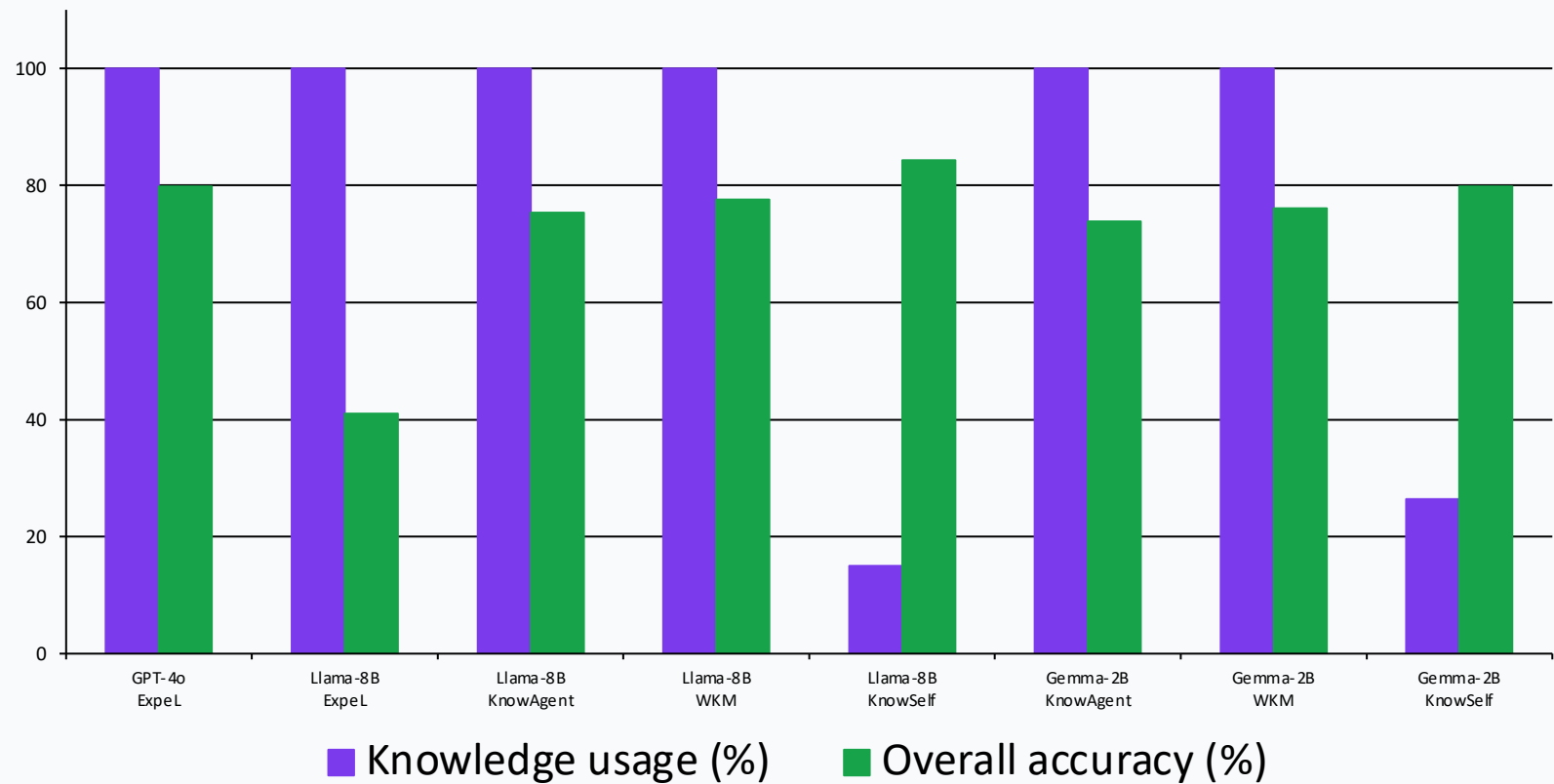
Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Source table reproduced from source brief. Bold-tinted rows mark KnowSelf; lavender rows mark 100% knowledge methods.

Minimal Knowledge, Maximum Impact

KnowSelf trades blanket knowledge injection for selective queries while matching or exceeding full-knowledge baselines.

Knowledge usage vs. ALFWorld overall accuracy



Efficiency readout

- Llama-8B KnowSelf uses **15.01%** knowledge and reaches **84.33%**.
- Gemma-2B KnowSelf uses **26.41%** knowledge and reaches **79.85%**.
- ExpeL / KnowAgent / WKM apply **100%** knowledge but do not consistently improve accuracy.
- Scaling down remains viable: Gemma-2B KnowSelf matches GPT-4o ExpeL overall.

Key claim: selective retrieval improves the accuracy-cost frontier.

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

KnowSelf converts planning from indiscriminate resource use into situational metacognitive control.

1

Flood irrigation is wasteful

Blindly injecting trajectories, feedback, or domain knowledge is inefficient and can be suboptimal.

2

Special tokens operationalize self-awareness

Heuristic labels on self-explored trajectories teach agents when to act, reflect, or retrieve.

3

Strong empirical gains

Llama-8B KnowSelf reaches 84.33%, above GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%).

4

Knowledge use drops sharply

Only 15.01–26.41% of steps use external knowledge vs. 100% for ExpeL / KnowAgent / WKM.

5

Scales across model sizes

Gemma-2B KnowSelf reaches 79.85%, matching GPT-4o ExpeL and exceeding Gemma-2B baselines.

Future directions: richer self-awareness criteria, broader planning domains, and tighter calibration of when retrieval is worth its cost.